



**Pakistan Institute Of Engineering And
Applied Sciences, PIEAS, Islamabad**

1 SEMESTER

COMPLEX ENGINEERING PROBLEM

An academic endeavor rooted in electrical engineering principles, this report presents a comprehensive solution to residential power continuity using simulation tools, analytical rigor, and system-level thinking.

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EE-111 CIRCUIT ANALYSIS 1

COMPLEX ENGINEERING PROBLEM

Design and Simulation of a Room-Wise UPS-Integrated
Residential Power System

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| CEP

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Abstract:

This report presents the design, analysis, and simulation of a room-wise **UPS-integrated** residential electrical system, developed as part of the Complex Engineering Problem (CEP) for the Circuit Analysis course. The project aimed to ensure continuous and reliable power delivery to essential household appliances by integrating an Uninterruptible Power Supply (UPS) into the wiring design. The modeled house comprised multiple rooms, bedrooms, kitchen, living room, guest room, garage, dining room, courtyard, and bathrooms, each equipped with typical electrical loads such as fans, LED bulbs, refrigerators, washing machines, computers, and other appliances. Using **MATLAB**, electrical parameters including voltage, current, resistance, and power were calculated component-wise, room-wise, and for the entire house. The total connected load was determined to be **8945 W**, with UPS-supported essential devices consuming approximately **1395 W**. **Multisim 14.2** simulations validated circuit behavior, switching logic, and load distribution under both main supply and backup conditions. The results confirmed that the designed system can reliably support critical low-power appliances during outages, while maintaining safety and efficiency through proper breaker selection and load management. The project also highlighted potential sources of error such as simplified assumptions, neglecting wire resistance, and DC-based modeling, with recommended precautions including AC-based simulations and real-world validation.

1. Objectives:

- Learned how to design house wiring systems using practical techniques introduced through this CEP.
- Developed hands-on understanding of circuit analysis by applying theoretical concepts to real-world scenarios.
- Gained practical experience with modern engineering tools like Multisim and MATLAB, which are essential for our future careers in Electrical Engineering.
- Learned how to test and validate circuit designs and theoretical models using simulation software.
- Built strong foundational concepts related to real-time engineering problems, helping us become more familiar with the professional field of Electrical Engineering.

2. Introduction:

The Complex Engineering Problem (CEP) undertaken as part of our Electrical Engineering program at PIEAS was designed to apply theoretical knowledge to practical scenarios in residential electrical systems. The project focused on home wiring design, circuit analysis, load management, and UPS integration, enabling us to simulate and analyze real-world conditions that engineers encounter in everyday practice. The project incorporated a range of household appliances and devices such as fans, LED bulbs, AC inverters, televisions, Wi-Fi routers, computers, microwave ovens, refrigerators, exhaust fans, door bells, water pumps, washing machines, and irons. By analyzing these components, we were able to evaluate load distribution, power consumption, and system reliability. This CEP not only strengthened our technical foundation but also enhanced our ability to collaborate, manage resources, and apply engineering solutions to complex real-life problems. It served as a bridge between classroom learning and practical application, preparing us for future challenges in the field of Electrical Engineering.

Resources:

To achieve these objectives, we utilized the following modern engineering tools:

- Matlab
- Multisim
- Canva

MATLAB:

I have used MATLAB for performing electrical calculations such as voltage, current, power, and resistance across individual components and rooms. MATLAB enabled precise analysis and automation of formulas based on circuit parameters.

MULTISIM:

I have used MULTISIM for simulating the complete home wiring circuit, including UPS integration. Multisim helped visualize circuit behavior, validate switching logic, and test load performance under both main supply and backup conditions.

Canva:

I used **Canva** to design professional layouts and visual representations of the home wiring project. Canva allowed me to create clear, visually appealing diagrams, flowcharts, and presentation slides that complemented the technical analysis performed in MATLAB and Multisim. By combining engineering data with effective visual design, Canva helped communicate complex information in a simple and impactful way.

Components:

I have used the following components

- Fan
- LED Bulb
- AC Inverter
- TV
- Wi-Fi Router
- Computer
- Microwave Oven
- Fridge
- Exhaust Fan
- Door Bell
- Water Pump
- Washing Machine
- Iron

3. Analysis:

This CEP includes the wiring of a house using modern tools like MULTISIM.

In this our Circuit includes:

- 1 Master Bedrooms
- 2 Bedrooms
- 1 Kitchen
- 1 Living room
- 1 Guest room
- 1 Grage
- 1 Dining Room
- 1 Courtyard
- 1 Entrance
- 4 Bathrooms

All bedrooms, kitchen, living room, guest room, Dining room, Grage, Courtyard, Bathrooms and Entrance of house are connected in parallel combinations of resistors so that all rooms gain equal voltage from the source. A voltage of 230 V is taken as a standard because in Pakistan the standard AC voltage source is 230 V and having the frequency of 50 Hz.

A.C. source was not used to make our circuit simple which saved us from the problem of calculating phase of A.C. current which also might change in daily routine. So, an independent D.C. source was used instead. The appliances were symbolized by their resistances which were calculated by using power formula. The powers of home appliances were in accordance with power availability in national market. The capacitors and inductors are also a part of circuit but to make a simple home circuit as asked by complex engineering problem, they were not added so. Current across each device was also calculated and in the final perspective power absorbed and power delivered were compared

The formula used for calculating the current passing through the loads are

by ohms law

$$V = IR$$

from here I

$$I = \frac{V}{R}$$

The formulas used for calculating the resistance of loads are

as,

$$P = \frac{V^2}{R}$$

so R is,

$$R = \frac{V^2}{P}$$

These formulas were used in MATLAB for calculations.

4. House Diagram:

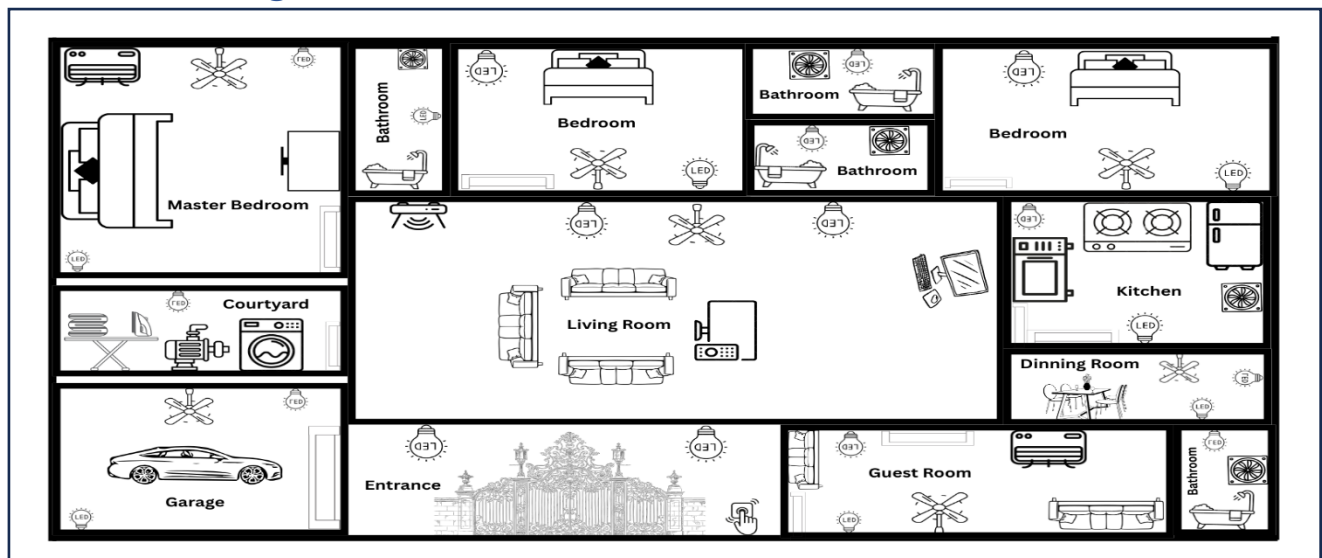


Figure 1: House Diagram by Canva

5. Components Power Rating:

Table 1: Components with their power ratings

Components	Power rating (W)
Fan	100
LED Bulb	15
AC Inverter	1500
Fridge	800
TV	65
Exhausts Fan	40
Wi-Fi Router	15
Computer	150
Microwave Oven	800
Door Bell	5
Washing Machine	600
Water Pump	1200
Iron	1000
UPS	2000

6. Calculations Using MATLAB:

In Pakistan, the standard electrical voltage is between 220V and 240V; therefore, we used a **230V** source for our project. We then used MATLAB to calculate the resistance and current across every component in the circuit. The calculations are broken down by component, then room by room, then for the total house load, followed by the UPS specifications, and are presented in the sections below:

Component Wise:

FAN

```
>> % Fan
>> format bank
>> V = 230;
>> P_Fan = 100;
>> R_Fan = V ^ 2 / P_Fan

R_Fan =

    529.00

>> I_Fan = P_Fan / V

I_Fan =

    0.43
```

Figure 2: MATLAB calculation of Fan

LED BULB

```
>> % LED Bulb
>> format bank
>> V = 230;
>> P_Bulb = 15;
>> R_Bulb = V ^ 2 / P_Bulb

R_Bulb =

   3526.67

>> I_Bulb = P_Bulb / V

I_Bulb =

    0.07
```

Figure 3: MATLAB calculation of LED Bulb

AC INVERTER

```
>> % AC Inverter
>> format bank
>> V = 230;
>> P_Inverter= 1500;
>> R_Inverter = V ^ 2 / P_Inverter

R_Inverter =

    35.27

>> I_Inverter = P_Inverter / V

I_Inverter =

    6.52
```

Figure 4: MATLAB calculation of AC Inverter

FRIDGE

```
>> % Fridge
>> format bank
>> V = 230;
>> P_Fridge= 800;
>> R_Fridge = V ^ 2 / P_Fridge

R_Fridge =

    66.12

>> I_Fridge = P_Fridge / V

I_Fridge =

    3.48
```

Figure 5: MATLAB calculation of Fridge

TV

```
>> % TV
>> format bank
>> V = 230;
>> P_TV= 65;
>> R_TV = V ^ 2 / P_TV

R_TV =

    813.85

>> I_TV = P_TV / V

I_TV =

    0.28
```

Figure 6: MATLAB calculation of TV

EXHAUST FAN

```
>> % Exhaust Fan
>> format bank
>> V = 230;
>> P_Exhaust= 40;
>> R_Exhaust = V ^ 2 / P_Exhaust

R_Exhaust =

    1322.50

>> I_Exhaust = P_Exhaust / V

I_Exhaust =

    0.17
```

Figure 7: MATLAB calculation of Exhaust Fan

WI-FI ROUTER

```
>> % Wi-Fi Router
>> format bank
>> V = 230;
>> P_Router= 15;
>> R_Router = V ^ 2 / P_Router

R_Router =

    3526.67

>> I_Router = P_Router / V

I_Router =

    0.07
```

Figure 8: MATLAB calculation of Wi-Fi Router

COMPUTER

```
>> % Computer
>> format bank
>> V = 230;
>> P_Computer= 150;
>> R_Computer = V ^ 2 / P_Computer

R_Computer =

    352.67

>> I_Computer = P_Computer / V

I_Computer =

    0.65
```

Figure 9: MATLAB calculation of Computer

MICROWAVE OVEN

```
>> % Microwave Oven
>> format bank
>> V = 230;
>> P_Oven= 800;
>> R_Oven = V ^ 2 / P_Oven

R_Oven =

    66.12

>> I_Oven = P_Oven / V

I_Oven =

    3.48
```

Figure 10: MATLAB calculation of Microwave Oven

DOOR BELL

```
>> % Door Bell
>> format bank
>> V = 230;
>> P_Bell= 5;
>> R_Bell = V ^ 2 / P_Bell

R_Bell =

   10580.00

>> I_Bell = P_Bell / V

I_Bell =

    0.02
```

Figure 11: MATLAB calculation of Door Bell

WASHING MACHINE

```
>> % Washing Machine
>> format bank
>> V = 230;
>> P_Machine= 600;
>> R_Machine = V ^ 2 / P_Machine

R_Machine =

    88.17

>> I_Machine = P_Machine / V

I_Machine =

    2.61
```

Figure 12: MATLAB calculation of Washing Machine

WATER PUMP

```
>> % Water Pump
>> format bank
>> V = 230;
>> P_Pump= 1200;
>> R_Pump = V ^ 2 / P_Pump

R_Pump =

    44.08

>> I_Pump = P_Pump / V

I_Pump =

    5.22
```

Figure 13: MATLAB calculation of Water Pump

IRON

```
>> % Iron
>> format bank
>> V = 230;
>> P_Iron = 1000;
>> R_Iron = V ^ 2 / P_Iron

R_Iron =

    52.90

>> I_Iron = P_Iron / V

I_Iron =

    4.35
```

Figure 14: MATLAB calculation of Iron

Room Wise:

MASTER BEDROOM

```
>> % Master Bedroom
>> format bank
>> I_M_Bedroom = I_Inverter + (2*I_Bulb) + I_Fan + I_TV

I_M_Bedroom =

    7.37

>> P_Absorbed1 = P_Inverter + (2*P_Bulb) + P_Fan + P_TV

P_Absorbed1 =

   1695.00

>> P_Delivered1 = V * I_M_Bedroom

P_Delivered1 =

   1695.00
```

Figure 15: MATLAB calculation of Master Bedroom

BEDROOM

```
>> % Bedroom
>> format bank
>> I_Bedroom = (2 * I_Bulb) + I_Fan

I_Bedroom =

    0.57

>> P_Absorbed2 = (2 * P_Bulb) + P_Fan

P_Absorbed2 =

   130.00

>> P_Delivered2 = V * I_Bedroom

P_Delivered2 =

   130.00
```

Figure 16: MATLAB calculation of Bedroom

GUEST ROOM

```
>> % Guest Room
>> format bank
>> I_G_Room = I_Inverter + (2 * I_Bulb) + I_Fan

I_G_Room =

    7.09

>> P_Absorbed3 = P_Inverter + (2 * P_Bulb) + P_Fan

P_Absorbed3 =

   1630.00

>> P_Delivered3 = V * I_G_Room

P_Delivered3 =

   1630.00
```

Figure 17: MATLAB calculation of Guest Room

LIVING ROOM

```
>> % Living Room
>> format bank
>> I_L_Room = I_Fan + I_TV + I_Router + I_Computer + (2 * I_Bulb)

I_L_Room =

    1.57

>> P_Absorbed4 = P_Fan + P_TV + P_Router + P_Computer + (2 * P_Bulb)

P_Absorbed4 =

   360.00

>> P_Delivered4 = V * I_L_Room

P_Delivered4 =

   360.00
```

Figure 18: MATLAB calculation of Living Room

KITCHEN

```
>> % Kitchen
>> format bank
>> I_Kitchen = I_Oven + I_Fridge + (2 * I_Bulb) + I_Exhaust

I_Kitchen =

    7.26

>> P_Absorbed5 = P_Oven + P_Fridge + (2 * P_Bulb) + P_Exhaust

P_Absorbed5 =

   1670.00

>> P_Delivered5 = V * I_Kitchen

P_Delivered5 =

   1670.00
```

Figure 19: MATLAB calculation of Kitchen

BATHROOM

```
>> % Bathroom
>> format bank
>> I_Bath = I_Exhaust + I_Bulb

I_Bath =

    0.24

>> P_Absorbed6 = P_Exhaust + P_Bulb

P_Absorbed6 =

    55.00

>> P_Delivered6 = V * I_Bath

P_Delivered6 =

    55.00
```

Figure 20: MATLAB calculation of Bathroom

DINING ROOM

```
>> % Dinning Room
>> format bank
>> I_Dinning = I_Fan + (2 * I_Bulb)

I_Dinning =

    0.57

>> P_Absorbed7 = P_Fan + (2 * P_Bulb)

P_Absorbed7 =

   130.00

>> P_Delivered7 = V * I_Dinning

P_Delivered7 =

   130.00
```

Figure 21: MATLAB calculation of Dining Room

COURTYARD

```
>> % Courtyard
>> format bank
>> I_Courtyard = I_Pump + I_Bulb + I_Machine + I_Iron

I_Courtyard =

   12.24

>>
>> P_Absorbed8 = P_Pump + P_Bulb + P_Machine + P_Iron

P_Absorbed8 =

  2815.00

>> P_Delivered8 = V * I_Courtyard

P_Delivered8 =

  2815.00
```

Figure 22: MATLAB calculation of Courtyard

GARAGE

```
>> % Garage
>> format bank
>> I_Garage = (2 * I_Bulb) + I_Fan

I_Garage =

    0.57

>> P_Absorbed9 = (2 * P_Bulb) + P_Fan

P_Absorbed9 =

   130.00

>> P_Delivered9 = V * I_Garage

P_Delivered9 =

   130.00
```

Figure 23: MATLAB calculation of Garage

ENTRANCE

```
>> % Entrance
>> format bank
>> I_Entrance = (2 * I_Bulb) + I_Bell

I_Entrance =

    0.15

>> P_Absorbed10 = (2 * P_Bulb) + P_Bell

P_Absorbed10 =

    35.00

>> P_Delivered10 = V * I_Entrance

P_Delivered10 =

    35.00
```

Figure 24: MATLAB calculation of Entrance

Total House:

TOTAL HOUSE

```
>> % Total House
>> format bank
>> I_Total_House1 = I_M_Bedroom + (2 * I_Bedroom) + I_G_Room + I_L_Room

I_Total_House1 =

   17.15

>> I_Total_House2 = I_Kitchen + (4 * I_Bath) + I_Dinning

I_Total_House2 =

    8.78

>> I_Total_House3 = I_Courtyard + I_Garage + I_Entrance

I_Total_House3 =

   12.96

>> I_Total_House = I_Total_House1 + I_Total_House2 + I_Total_House3

I_Total_House =

   38.89
```

```

>> P_Total_House_Absorbed1 = P_Absorbed1 + (2 * P_Absorbed2) + P_Absorbed3 + P_Absorbed4

P_Total_House_Absorbed1 =

    3945.00

>> P_Total_House_Absorbed2 = P_Absorbed5 + (4 * P_Absorbed6) + P_Absorbed7

P_Total_House_Absorbed2 =

    2020.00

>> P_Total_House_Absorbed3 = P_Absorbed8 + P_Absorbed9 + P_Absorbed10

P_Total_House_Absorbed3 =

    2980.00

>> P_Total_House_Absorbed = P_Total_House_Absorbed1 + P_Total_House_Absorbed2 + P_Total_House_Absorbed3

P_Total_House_Absorbed =

    8945.00

```

```

>> P_Total_House_Delivered = V * I_Total_House

P_Total_House_Delivered =

    8945.00

```

Figure 25: MATLAB calculation of Total House

UPS:

UPS

```

>> % UPS
>> format bank
>> I_UPS_House = (23 * I_Bulb) + (7 * I_Fan) + I_Router + (2 * I_TV) + (5 * I_Exhaust) + I_Bell

I_UPS_House =

    6.07

```



```

>> P_UPS_House_Absorbed = (23 * P_Bulb) + (7 * P_Fan) + P_Router + (2 * P_TV) + (5 * P_Exhaust) + P_Bell

P_UPS_House_Absorbed =

    1395.00

>> P_UPS_House_Delivered = V * I_UPS_House

P_UPS_House_Delivered =

    1395.00

```

Figure 26: MATLAB calculation of UPS

MAIN SOURCE

```

>> % Main Source
>> format bank
>> I_Main_Source = (2 * I_Inverter) + I_Fridge + I_Computer + I_Oven + I_Machine + I_Pump + I_Iron

I_Main_Source =

    32.83

>> P_Main_Source_Absorbed = (2 * P_Inverter) + P_Fridge + P_Computer + P_Oven + P_Machine + P_Pump + P_Iron

P_Main_Source_Absorbed =

    7750.00

>> P_Main_Source_Delivired = V * I_Main_Source

P_Main_Source_Delivired =

    7550.90

```

Figure 27: MATLAB calculation of UPS

These calculations were fundamental in dimensioning the circuit breakers and selecting appropriate wire gauges to ensure safety and efficiency throughout the electrical system.

Results Of MATLAB:

Table 2: Component Wise Results of MATLAB

Components	Power (W)	Resistance (Ω)	Current (A)
Fan	100	529	0.43
LED Bulb	15	3526.27	0.07
AC Inverter	1500	35.27	6.52
Fridge	800	66.12	3.48
TV	65	813.85	0.28
Exhausts Fan	40	1322.50	0.17
Wi-Fi Router	15	3526.67	0.07
Computer	150	352.67	0.65
Microwave Oven	800	66.12	3.48
Door Bell	5	10580	0.02
Washing Machine	600	88.17	2.61
Water Pump	1200	44.08	5.22
Iron	1000	52.90	4.35

Table 3: Room Wise Results of MATLAB

Room	Current (A)	Power Absorbed (W)	Power Delivered (W)
Master Bedroom	7.37	1695	1695
Bedroom	0.57	130	130
Guest Room	7.09	1630	1630
Living Room	1.57	360	360
Kitchen	7.26	1670	1670
Bathroom	0.24	55	55
Dinning Room	0.57	130	130
Courtyard	12.24	2815	2815
Garage	0.57	130	130
Entrance	0.15	35	35

Table 4: Total House Results of MATLAB

Total House	
Current	38.89 A
Power Absorbed	8945 W
Power Delivered	8945 W

Table 5: UPS Results of MATLAB

UPS House	
Current	6.07 A
Power Absorbed	1395 W
Power Delivered	1395 W

Table 6: Main Source Results of MATLAB

Main Source House	
Current	32.83 A
Power Absorbed	7550 W
Power Delivered	7550 W

7. UPS Selection:

After calculating the total connected load of household appliances, the required power usage was found to be 1395W. To ensure reliable operation and account for efficiency losses, surge currents, and future expansion, we selected a 2000W UPS for installation in the house.

This choice provides the following benefits:

- **Safety Margin:**

The UPS capacity exceeds the calculated load, preventing overload and ensuring stable performance.

- **Surge Handling:**

High-wattage appliances such as refrigerators, washing machines, and microwaves often draw extra current at startup. A 2000W UPS can handle these surges without tripping.

- **Efficiency:**

Operating the UPS below its maximum rating improves efficiency and prolongs its lifespan.

- **Future Flexibility:**

Additional small appliances can be connected later without exceeding the UPS capacity.

8. Circuit Diagrams:

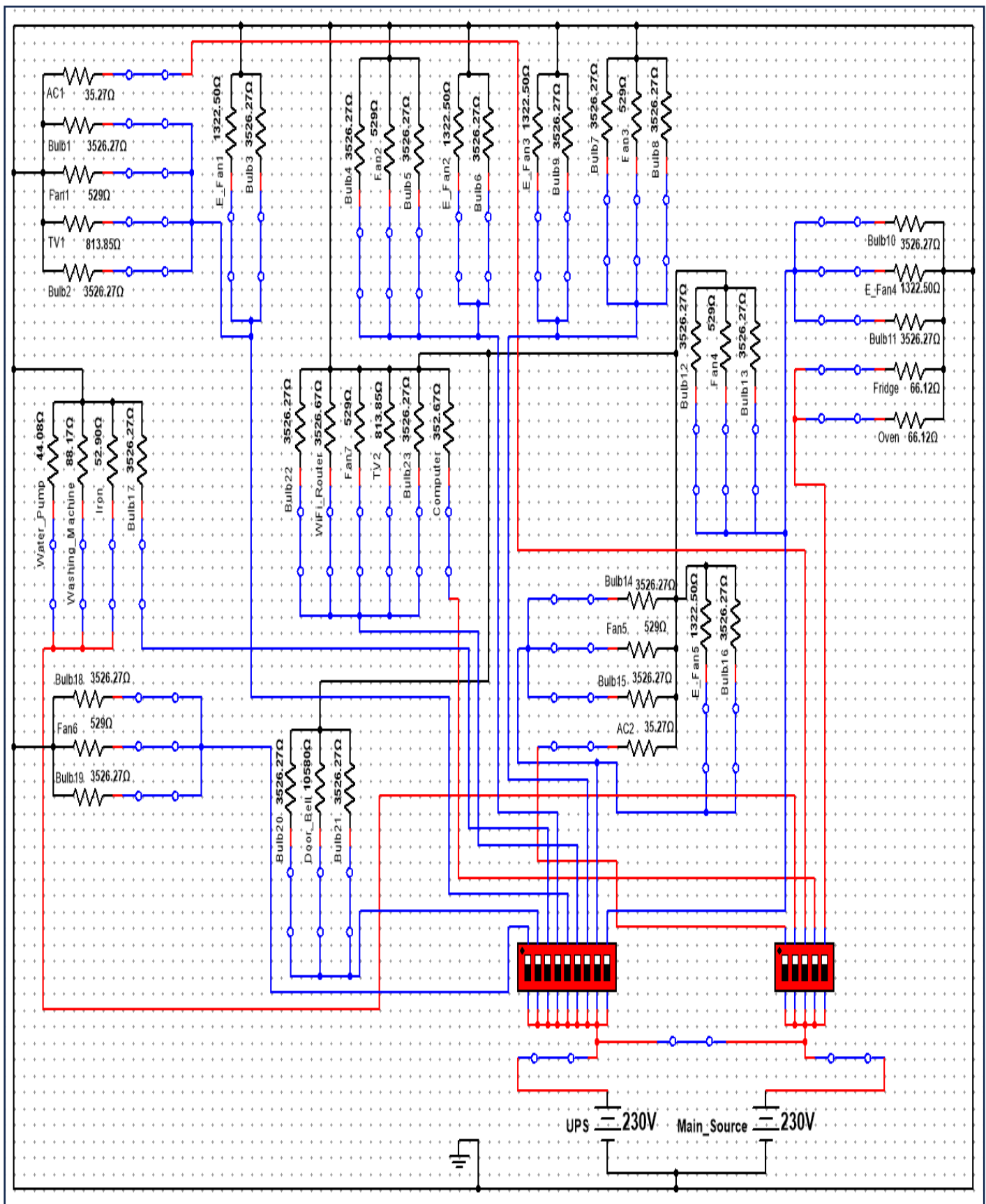


Figure 28: House Circuit Design, Multisim 14.2

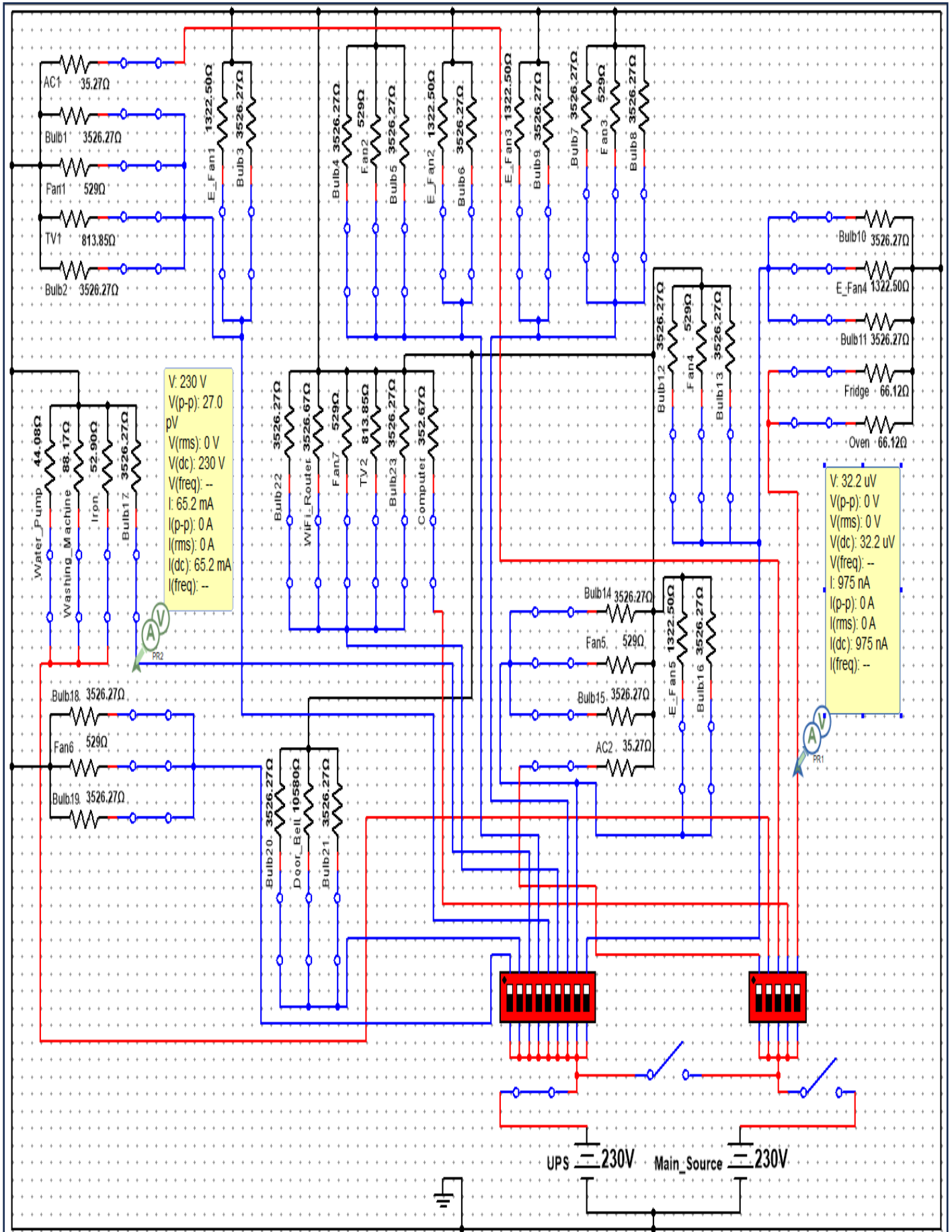


Figure 29: UPS connected House Circuit Design, Multisim 14.2

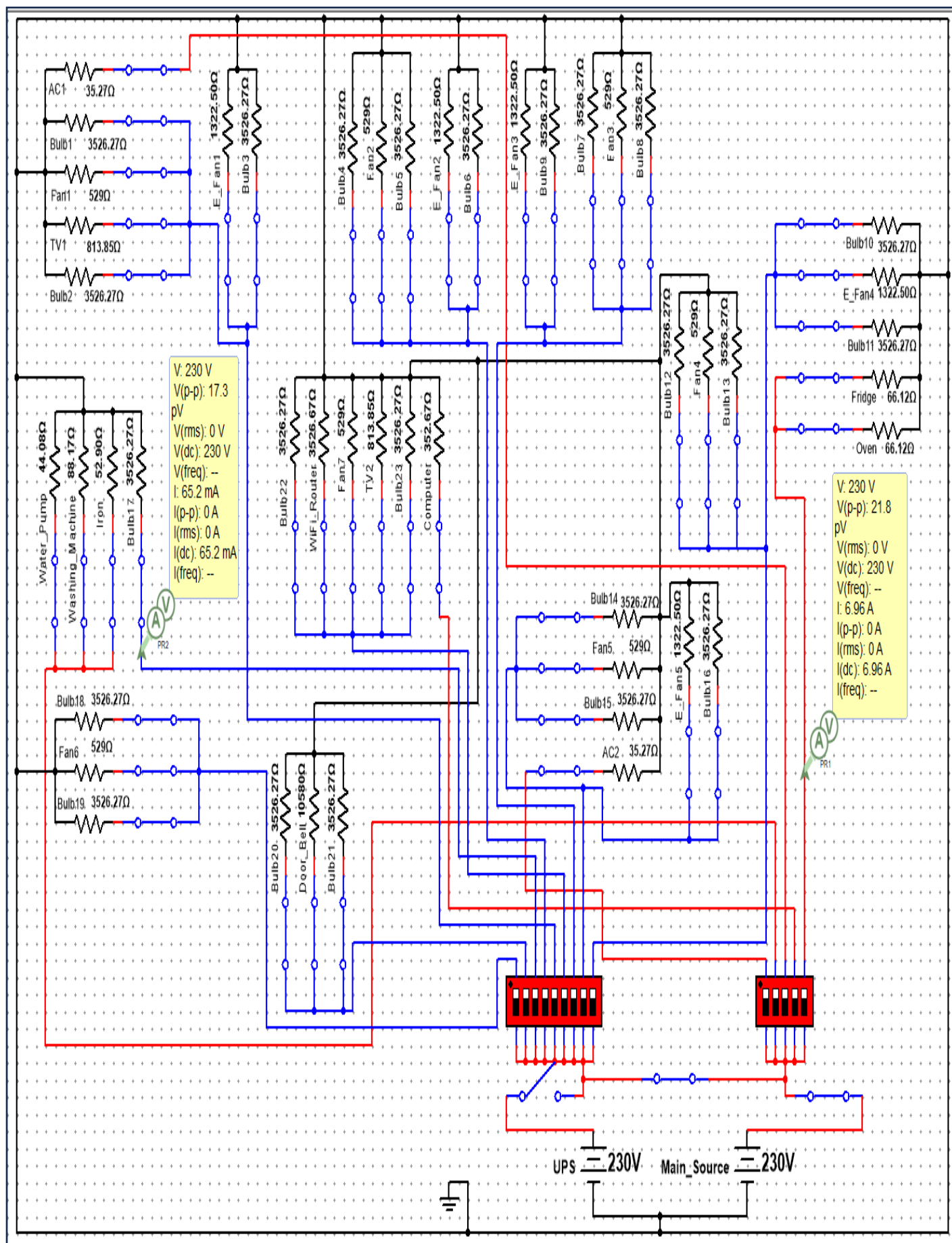


Figure 30: Main source connected House Circuit Design, Multisim 14.2

9. Discussion:

The completion of our Complex Engineering Problem (CEP) provided valuable insights into the design and operation of residential electrical systems. Using **MATLAB**, we performed detailed calculations of voltage, current, power, and resistance to evaluate load distribution across household appliances. This analytical stage ensured accuracy in theoretical modeling, while **NI Multisim 14.2** allowed us to simulate the complete wiring circuit, visualize switching logic, and test performance under both main supply and UPS backup conditions. The inclusion of appliances such as fans, refrigerators, washing machines, and microwave ovens highlighted the practical challenges of balancing efficiency, safety, and reliability in real-world scenarios. Protective devices like circuit breakers and fuses were emphasized as critical safeguards against overloads and short circuits, ensuring system stability. Additionally, **Canva** was used to design professional layouts and diagrams, making complex data easier to interpret and present.

Errors and Precautions:

However, it is important to acknowledge that **real-time electrical calculations are rarely 100% accurate** due to practical limitations and assumptions.

Sources of Error:

Several sources of error were identified during the project:

- **Assumptions in Theoretical Models:** Simplified assumptions often fail to capture real-world complexities.
- **Use of DC Voltage Instead of AC:** For convenience, DC values were used in calculations, whereas household wiring operates on AC, leading to deviations.
- **Full Power Utilization:** Appliances were assumed to consume their full rated power continuously, though actual usage varies with operating conditions.
- **Ignoring Wire Resistance:** Voltage drops and power losses due to conductor resistance were not included in ideal calculations.
- **Circuit Breaker Behavior:** Protective tripping of breakers under simultaneous high loads was not modeled but remains critical in practice.

Precautions to Minimize Errors:

To minimize these errors, several precautions are recommended:

- Employ **AC-based simulation models** to better reflect household conditions.
- Include **wire resistance** in calculations to account for voltage drops and losses.
- Model **load variability** using average or dynamic consumption values rather than peak ratings.
- Simulate **circuit protection devices** such as breakers and fuses to understand system behavior under overload conditions.
- Validate simulation results with **real measurements** wherever possible to refine accuracy.

This project not only strengthened our technical foundation but also emphasized the importance of recognizing limitations in theoretical models and simulations. By combining analytical tools, simulation software, and professional presentation design, we bridged the gap between classroom learning and practical application. Equally significant was the teamwork among the

three group members, which enabled effective division of tasks, troubleshooting, and integration of diverse perspectives. Overall, the CEP enhanced our ability to approach complex engineering problems with precision, innovation, and a strong awareness of practical constraints.

10. Conclusion:

The Complex Engineering Problem (CEP) provided a comprehensive opportunity to apply theoretical knowledge of Electrical Engineering to a practical residential system. By combining analytical calculations in MATLAB, circuit simulations in Multisim, and professional visualization through Canva, we successfully designed and evaluated a home wiring layout with integrated backup power. The inclusion of diverse household appliances allowed us to study load distribution, safety mechanisms, and system reliability in detail. While the project highlighted the effectiveness of modern tools in bridging theory and practice, it also revealed the limitations of idealized models. Assumptions such as continuous full power utilization, neglecting wire resistance, and simplified breaker behavior introduced potential errors. Precautions such as adopting AC-based models, simulating protective devices, and validating results with real measurements are essential for improving accuracy in future work. Beyond technical outcomes, the CEP emphasized the importance of team collaboration, problem-solving, and professional presentation. Working as a group of three enabled efficient task division, collective troubleshooting, and integration of diverse perspectives. Ultimately, this project strengthened our technical foundation, enhanced our confidence in tackling real-world engineering challenges, and prepared us to approach future problems with innovation, precision, and responsibility. Overall, the CEP bridged the gap between classroom learning and practical application. It strengthened our technical foundation, enhanced our problem-solving skills, and prepared us to address future engineering challenges with confidence, innovation, and professionalism.

References:

- MATLAB Documentation – <https://www.mathworks.com/help>
- NI Multisim User Guide – <https://www.ni.com/en-us/support/documentation.html>
- Electrical Engineering Fundamentals by Vincent Del Toro
- Residential Wiring Guide – Pakistan Electric Rules & Standards
- Datasheets for appliances (AC, LED, Fan, TV, etc.) from manufacturer websites
- Online tutorials and simulation examples from YouTube and engineering forums
- Canva – <https://www.canva.com> (used for layout and presentation design)